Loan Approval Prediction

Complete Project Report

Model: XGBoost Classifier (Tuned via GridSearchCV)

Dataset: loan.csv (4,269 samples, 13 columns)

Target: Loan Approval (Binary Classification)

Generated automatically from project artifacts

1. Libraries Used

Library	Purpose	Key Functions
pandas	Data manipulation & CSV loading	pd.read_csv, DataFrame ops
numpy	Numerical computing	Array operations
matplotlib.pyplot	Data visualization	Histograms, boxplots, scatter
seaborn	Statistical visualization	Heatmaps, countplots, violinplots
sklearn.preprocessing	Feature engineering	LabelEncoder, StandardScaler
sklearn.model_selection	Model selection	train_test_split, GridSearchCV, RandomizedSearchCV
sklearn.linear_model	Baseline model	LogisticRegression
sklearn.tree	Decision tree model	DecisionTreeClassifier
sklearn.ensemble	Ensemble model	RandomForestClassifier
xgboost	Gradient boosting	XGBClassifier (final model)
sklearn.metrics	Evaluation metrics	accuracy, precision, recall, f1, roc_auc, confusion_matrix
joblib	Model serialization	dump/load .pkl files
streamlit	Web app deployment	Interactive prediction UI

2. Dataset Overview

The dataset (loan.csv) contains 4269 rows and 13 columns. The target variable is 'loan_status' with two classes: Approved (2656 samples, 62.2%) and Rejected (1613 samples, 37.8%).

2.1 Feature Descriptions

Column	Dtype	Description
loan_id	int64	Unique loan identifier (dropped before training)
no_of_dependents	int64	Number of dependents of the applicant
education	object	Graduate / Not Graduate
self_employed	object	Yes / No
income_annum	int64	Annual income in INR
loan_amount	int64	Requested loan amount in INR
loan_term	int64	Loan term in months
cibil_score	int64	Credit score (300-900)
residential_assets_value	int64	Value of residential assets
commercial_assets_value	int64	Value of commercial assets
luxury_assets_value	int64	Value of luxury assets
bank_asset_value	int64	Value of bank assets
loan_status	object	Approved / Rejected (target)

2.2 Derived Features (Created During Feature Engineering)

Feature	Formula

total_assets	Sum of all 4 asset columns (computed after scaling)
debt_income_ratio	loan_amount / (income_annum + 1) (computed after scaling)

3. Preprocessing Pipeline

3.1 Label Encoding (Categorical Features)

Column	Mapping	Method
education	Graduate = 0, Not Graduate = 1	LabelEncoder (alphabetical)
self_employed	No = 0, Yes = 1	LabelEncoder (alphabetical)
loan_status	Approved = 0, Rejected = 1	LabelEncoder (alphabetical)

3.2 Standard Scaling (Numerical Features)

StandardScaler was applied to 8 numerical columns. Formula: z = (x - mean) / std. The scaler was fit on the full dataset before train/test split.

Feature	Mean	Std Dev
income_annum	5,059,123.92	2,806,511.07
loan_amount	15,133,450.46	9,042,303.73
loan_term	10.90	5.71
cibil_score	599.94	172.41
residential_assets_value	7,472,616.54	6,502,874.81
commercial_assets_value	4,973,155.31	4,388,452.01
luxury_assets_value	15,126,305.93	9,102,687.34
bank_asset_value	4,976,692.43	3,249,804.61

3.3 Train/Test Split

Data was split using train_test_split with:
- test_size = 0.2 (80% train, 20% test)
- random_state = 42
- stratify = y (preserves class distribution)

Set	Size	Proportion
Training Set	3415 samples	80%
Test Set	853 samples	20%

4. Models Trained

4.1 Logistic Regression (Baseline)

Parameter	Value	Description
max_iter	1000	Maximum iterations for solver convergence
solver	lbfgs	Default optimizer
penalty	l2	Default regularization
C	1.0	Inverse regularization strength

4.2 Decision Tree Classifier

Parameter	Value	Description
criterion	gini	Split quality measure
max_depth	None	No depth limit (fully grown tree)
random_state	42	Reproducibility seed
splitter	best	Default split strategy
min_samples_split	2	Default minimum samples to split
min_samples_leaf	1	Default minimum samples per leaf

4.3 Random Forest Classifier

Parameter	Value	Description
n_estimators	100	Number of trees in the forest
criterion	gini	Split quality measure
max_depth	None	No depth limit
random_state	42	Reproducibility seed
n_jobs	-1	Default (single core, unless overridden)
bootstrap	True	Default bootstrap sampling

4.4 XGBoost Classifier (Default Parameters)

Parameter	Value	Description
objective	binary:logistic	Binary classification objective
eval_metric	logloss	Log loss evaluation metric
use_label_encoder	False	Disabled (deprecated feature)
n_estimators	100	Default number of boosting rounds
max_depth	6	Default maximum tree depth
learning_rate	0.3	Default step size shrinkage
random_state	42	Reproducibility seed

5. Hyperparameter Tuning

5.1 GridSearchCV Configuration

GridSearchCV was used with:
- scoring = 'f1'
- cv = 3 (3-fold cross-validation)
- n_jobs = -1 (parallel processing)
- verbose = 2

Parameter	Search Space	Count
n_estimators	[100, 200, 300]	3 values
max_depth	[3, 5, 7]	3 values
learning_rate	[0.01, 0.05, 0.1]	3 values
subsample	[0.8, 1.0]	2 values
colsample_bytree	[0.8, 1.0]	2 values

Total combinations: 3 x 3 x 3 x 2 x 2 = 108 (x 3 folds = 324 fits)

5.2 RandomizedSearchCV Configuration

RandomizedSearchCV was also used with:
- n_iter = 20 (20 random combinations)
- scoring = 'f1'
- cv = 3
- random_state = 42

Parameter	Search Space	Count
n_estimators	[100, 200, 300, 400, 500]	5 values
max_depth	[3, 4, 5, 6, 7, 8]	6 values
learning_rate	[0.01, 0.05, 0.1, 0.2]	4 values
subsample	[0.6, 0.8, 1.0]	3 values
colsample_bytree	[0.6, 0.8, 1.0]	3 values
gamma	[0, 1, 5]	3 values

5.3 Best Parameters (from GridSearchCV - used for final model)

Parameter	Best Value
n_estimators	100
max_depth	7
learning_rate	0.05
subsample	0.8
colsample_bytree	0.8
objective	binary:logistic
eval_metric	logloss

random_state	42
use_label_encoder	False

6. Final Model - Complete Hyperparameters

The final deployed model is an XGBClassifier trained with the best parameters from GridSearchCV. Below is the exhaustive list of every hyperparameter.

Hyperparameter	Value
base_score	None (default)
booster	None (default)
callbacks	None (default)
colsample_bylevel	None (default)
colsample_bynode	None (default)
colsample_bytree	0.8
device	None (default)
early_stopping_rounds	None (default)
enable_categorical	False
eval_metric	logloss
feature_types	None (default)
feature_weights	None (default)
gamma	None (default)
grow_policy	None (default)
importance_type	None (default)
interaction_constraints	None (default)
learning_rate	0.05
max_bin	None (default)
max_cat_threshold	None (default)
max_cat_to_onehot	None (default)
max_delta_step	None (default)
max_depth	7
max_leaves	None (default)
min_child_weight	None (default)
missing	nan
monotone_constraints	None (default)
multi_strategy	None (default)
n_estimators	100
n_jobs	None (default)
num_parallel_tree	None (default)
objective	binary:logistic
random_state	42
reg_alpha	None (default)

reg_lambda	None (default)
sampling_method	None (default)
scale_pos_weight	None (default)
subsample	0.8
tree_method	None (default)
use_label_encoder	False
validate_parameters	None (default)
verbosity	None (default)

7. Feature Importance (Final Model)

Feature importances from the final XGBoost model, measured by gain. CIBIL score dominates with 73.7% importance.

Rank	Feature	Importance	Percentage
1	cibil_score	0.737196	73.72%
2	loan_term	0.148272	14.83%
3	debt_income_ratio	0.018674	1.87%
4	loan_amount	0.015004	1.50%
5	income_annum	0.013439	1.34%
6	total_assets	0.010404	1.04%
7	no_of_dependents	0.009174	0.92%
8	commercial_assets_value	0.008924	0.89%
9	residential_assets_value	0.008510	0.85%
10	bank_asset_value	0.008462	0.85%
11	education	0.007938	0.79%
12	luxury_assets_value	0.007725	0.77%
13	self_employed	0.006278	0.63%

8. Model Evaluation Metrics

Below are the evaluation metrics reported in the notebook for each model on the 20% test set. Values are approximate from notebook outputs.

8.1 Metrics Comparison

Model	Config	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	Baseline	~89%	~89%	~93%	~91%	~96%
Decision Tree	Default	~98%	~98%	~98%	~98%	~98%
Random Forest	Default	~98%	~99%	~98%	~98%	~99%
XGBoost	Default	~98%	~98%	~98%	~98%	~99%
XGBoost (Tuned)	GridSearchCV	~98%	~98%	~99%	~98%	~99%

Note: All tree-based models achieved very similar high performance. The tuned XGBoost was selected as the final model based on GridSearchCV's best F1 score.

8.2 Key Observations

1. CIBIL Score is the dominant predictor (73.7% feature importance), acting as a near-binary classifier. Scores above 551 are almost always approved; below are almost always rejected.

2. Loan Term is the second most important feature (14.8%), with shorter terms slightly favoring approval.

3. All other features (income, assets, education, self-employment) have minimal individual impact on predictions (less than 2% each).

4. The model achieves ~98% accuracy, indicating strong separation in the data primarily driven by credit score (CIBIL).

9. Deployment

The model is deployed as a Streamlit web application (app.py) that:

1. Loads the saved model (loan_approval_model.pkl), scaler (scaler.pkl), and feature order (feature_order.pkl)
2. Accepts user inputs via an interactive form
3. Applies the same preprocessing pipeline: LabelEncoding, StandardScaler, derived features
4. Displays Approved/Rejected prediction with confidence percentage

Run command: streamlit run app.py

9.1 Saved Artifacts

File	Contents
loan_approval_model.pkl	Tuned XGBClassifier (final model)

feature_order.pkl	List of 13 feature names in training order
scaler.pkl	Fitted StandardScaler for 8 numerical columns
loan.csv	Original dataset (4,269 rows)